

Gaze cuing effect in action control: Yes, action matters!

ABSTRACT

One of the core components of non-verbal communication and joint attention is gaze cuing – our tendency to shift attention in the direction of another’s gaze. While extensively studied, little is known about whether and how internal states like the sense of agency influence sensitivity to gaze cues. In this study, we investigate whether control over action (action-outcome consistency) modulates gaze cuing effect (GCE) in a color discrimination task. Participants viewed a synthetic face with lateral (left/right) pupil motion, followed by the appearance of a colored target to the left/right of the face. During each trial, participants either passively observed the face stimulus onset (control condition), or actively initiated it by selecting a gaze direction and pressing a corresponding button. Crucially, in active trials, the gaze direction could be consistent or inconsistent with the chosen direction. The results showed larger in magnitude GCE in action-outcome consistent condition; moreover, time distribution analyses suggested that the GCE appeared early on, in contrast to control and action-outcome inconsistent conditions. Together, these data point towards a possible (facilitative) link between action control and social attentional orienting.

KEYWORDS

Gaze cuing, Action control, Sense of agency, Social cognition, Social attention, RT.

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Literature

Gaze Cueing and Attentional Orienting

Our environment is crowded with objects that often require further scrutiny, and attending to goal-relevant features of the environment is essential for effective functioning. However, attention is a finite resource and must be selectively deployed [1]. Consequently, people use various directional cues to focus their attention on important events and objects in the world. As highly social creatures, humans rely strongly on social cues like gaze direction, head orientation, or pointing gestures.

Reported at the end of the 20th century, Gaze cueing (GC) refers to the human tendency to shift attention in the direction of another person’s gaze [2], [3], [4], [5]. It serves as a fundamental social cue, allowing individuals to infer the attentional focus of others and thereby engage in joint attention—a core component of human social interaction [6], [7], [8]. As such, GC plays a central role in non-verbal and embodied communication, and is considered one of the foundational mechanisms supporting the development of theory of mind and the capacity to navigate complex social environments

[6], [7], [8]. Originally considered a reflexive mechanism [2], [3], [4], [5], GC has been widely used to investigate attentional orienting processes. It is most commonly studied using a variant of the Posner cueing paradigm [9], a classic attentional task in which a centrally presented cue (e.g., an arrow or a flash) precedes a target, and responses are typically faster when the cue validly predicts the target’s location.

In the GC version, the symbolic cue is replaced by a centrally presented face whose gaze shifts left or right. Even when gaze direction is non-predictive of the target’s location, participants respond faster to targets appearing in the gazed-at direction—a phenomenon known as the gaze cuing effect (GCE). This effect has been interpreted as evidence for the reflexive and automatic nature of gaze-induced attentional shifts [2], [3], [4], [5]. Later support for those findings come also from eye-tracking studies that show that gaze-congruent targets elicit faster saccadic movements [10], [11].

However, the reflexive nature of gaze cueing has since been questioned. A growing body of research suggests that, while gaze cueing may be fast and seemingly automatic, it is not purely reflexive but is modulated by contextual, emotional, and social factors, as well as higher-order cognitive processes (e.g., [35]; and for a review, see [12]). These findings indicate that gaze cueing is deeply intertwined with social meaning and cognitive mechanisms, highlighting its dynamic and flexible nature and setting the stage for further exploration of its modulatory factors.

Modulators of Gaze Cueing

Modulators of the GCE are highly researched and can be generally grouped into three broad categories: (1) physical characteristics of the cueing agent, (2) internal states of the observer, and (3) interactions between these factors [12].

Physical characteristics of the cueing face include features such as facial expression, dominance, gender or sense of social status. A substantial body of research supports the idea that those characteristics can influence GCE, demonstrating, for instance, that individuals exhibit stronger GCEs when viewing high-status faces [13], more dominant faces [14], [15] or familiar faces that participants have previously engaged in joint attention with [16], [17].

Emotional expressions have also been considered as modulators of GC, especially under specific conditions. For instance, a greater GCE is induced by happy faces of female actors [18]; or faces that previously smiled to the participants (only in low Autism-Spectrum Quotient individuals) [19]; Greater GCE is also shown for angry faces, but only when participants were under high cognitive load [20]; or for happy and fearful faces after oxytocin administration [21].

The internal differences of the observer are thought to be another cluster of factors that have their role as modulatory factors. Traits such as elevated anxiety [22], [23], higher levels of fear [24], or

phobias combined with higher levels of anxiety traits [25] have been linked to stronger GCEs, as have low self-esteem and heightened social needs [26]. More stable personality traits like intro- and extraversion also interact with emotional expression with introverts responding strongly to neutral and happy faces, while extraverts are showing higher sensitive to angry ones [27]. Gender may play a role as well, with stronger GCEs reported in females than males [28].

Beyond stable traits, transient internal states such as social acceptance or rejection [29], or primed feelings of power [30], have been shown to modulate the GCE—underlying GCE’s sensitivity to both permanent and temporary internal conditions. Interestingly, political temperament also appears to influence gaze cueing, with liberals showing more robust effects than conservatives [31].

As the reviewed studies illustrate, the gaze cueing effect (GCE) is rarely shaped by a single, isolated factor. Rather, it often results of a dynamic interaction between factors from both groups – the characteristics of the cueing face (e.g., expression, status, trustworthiness) and the traits of the observer (e.g., anxiety, social motivation). The interaction effects reflect how individual and social factors combine in shaping attentional responses – suggesting that gaze cueing is not merely a reflexive shift of attention, but a flexible, context-sensitive process.

Despite the large body of research, one internal state remains notably underexplored in the gaze cueing literature: the sense of agency (SoA). While prior work has largely emphasized stable personality traits, much less is known about how transient, moment-to-moment cognitive states—such as the subjective experience of control over one’s own actions—might modulate social attention. Investigating SoA thus represents a valuable opportunity to deepen our understanding of how volitional control dynamically interacts with mechanisms of gaze-based attentional orienting and non-verbal communication.

The Sense of Agency

The sense of agency (SoA) can be defined as the pre-reflective feeling of being the agent of one’s own actions [32], and the experience of controlling them and, through them, controlling events in the external world [33]. In other words, it is the feeling that "I am the one who is performing the action, hence I’m the one causing the effect." SoA is fundamental to motor control, self-awareness, and social interaction, where recognizing one’s own contributions is essential for coordinating joint actions [32]. As discussed in [34], SoA also supports the development of joint attention, a critical capacity for aligning visual focus and inferring others’ intentions (e.g., theory of mind - ToM).

Several theoretical frameworks have been proposed to explain how SoA arises. Predictive coding models suggest that agency stems from the match between predicted and actual sensory outcomes [20], [16]. Cue integration models propose that internal motor signals and external cues are combined and taken into account for the final judgment of agency (Pacherie, 2013, in [34]). Retrospective inference models, in their turn, argue that agency can also be inferred after the fact, based on contextual and temporal

information [33]. Despite their differences, all three models converge on a central claim: congruency/consistency between intended actions and observed outcomes enhances the experience of agency [34].

Importantly, recent research shows that SoA is not limited to motor actions like button presses. For instance, [9] demonstrated that people can experience agency over eye movements, particularly when gaze behavior leads to predictable outcomes such as color changes. Their study revealed that gaze–outcome contingency increased both explicit agency ratings and implicit ratings (measured by temporal binding effects), extending the research of SoA to the oculomotor domain.

Other studies highlight the social nature of agency. For instance, [29] found that mutual gaze enhances perceived agency in social contexts. Participants felt more in control of outcomes when their gaze was met by another’s, suggesting that social cues like eye contact modulate SoA—a finding that underlines the role of interpersonal feedback in agency judgments.

Together, these results position SoA as multimodal, context-sensitive, and deeply social state, extending action control from mere physical action into attentional and communicative domains.

Agency’s Influence on Perception and Attention

A growing body of evidence shows that SoA affects not only how actions are experienced, but also how sensory and attentional processes unfold. For example, the intentional binding effect—a compression in perceived time between an action and its effect—is a widely used implicit measure of agency [26]. Likewise, sensory attenuation refers to the phenomenon where self-generated stimuli (e.g., sounds, touches) are perceived as less intense, presumably due to predictive down-weighting [2], [31].

SoA also influences higher-level attention and perceptual salience. As summarized by Wen and Imamizu (2022), when individuals believe they control an object or event, their attention tends to be drawn more strongly to it, even when they are not instructed to look there. This effect elevates with higher levels of action–outcome congruency/consistency [30]. The effect remains present in social context as well – in joint activities, the distribution of attention is also shaped by shared agency. When people feel mutually responsible for an outcome, they align their attention with that of their partner to coordinate more effectively [30]. This mutual sense of control promotes synchronized gaze, enhances prediction of others’ behavior, and helps focus attention on task-relevant cues [34]. Overall, the studies support the notion that SoA prioritizes attention toward elements in the environment that are perceived as linked to one’s own or shared actions and goals.

In the social domain, agency and attention appear to interact closely. As mentioned above, Ulloa et al. (2019) [29] demonstrated that eye contact not only communicates engagement but also enhances SoA. Meanwhile, Gutzeit et al. (2024) [9] showed that even gaze-contingent effects—where simply looking in a direction causes an event—are sufficient to elicit SoA, particularly when the action–effect relationship is perceived as contingent and predictable.

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Considering the existing literature on both gaze cueing (GC) and the sense of agency (SoA), as well as the many points of intersection between them—such as their importance for attentional allocation, their susceptibility to social influences, and their major role for social cognition and functioning—it is logical to ask whether there is an overlap between these two mechanisms, and if so, what its nature might be.

Current Study

The present experiment is the first part of a larger research program investigating how different types of actions—ranging from bodily movements to minimal oculomotor shifts—interact with attentional and social-perceptual mechanisms such as gaze cueing.

Despite its theoretical significance and centrality to volitional behavior, the influence of the sense of agency on attentional processes—specifically gaze cueing—remains poorly understood. While prior research has shown that gaze behavior can influence SoA (e.g., mutual gaze enhancing agency; [29], [10]), the reverse relationship—whether agency modulates the perceptual and attentional impact of gaze cues—has not been directly tested. Our study addresses this gap by investigating whether SoA shapes the strength of GCE.

In this experiment, we operationalize SoA through intentional (action–outcome) consistency: whether the direction of a gaze shift aligns with the participant’s intention. Participants perform a color discrimination task in which a synthetic face with dynamic pupil motion precedes target onset. In active trials, participants choose a gaze direction, which then either is consistent or inconsistent with the resulting gaze direction. In passive (control) trials, gaze direction is externally “determined” and participants have no control over it. The gaze direction itself is either valid (target appears on the same side) or invalid (opposite side), allowing us to measure whether agency/action control modulates the behavioral impact of gaze cues. Finally, we have chosen discrimination between two very similar colors to make the task more difficult (hence, slow) in order to open some ‘space’ for the agency to penetrate the GCE.

This design allows us to examine whether internal states like SoA shape social attention. Drawing on predictive processing [35], [36] and cue integration frameworks (Pacherie, 2013, in [34]), as well as empirical findings showing that consistent action–outcome relationships enhance both agency and attentional salience ([9], [25]), we proposed that the effectiveness of gaze cues will vary systematically with perceived agency.

Specifically, we expected enhanced gaze cueing effect in action–outcome intentionally consistent condition, where participants’ chosen direction would follow by an intended gaze shift, reinforcing predictive consistency and strengthening SoA. By contrast, action–outcome intentionally inconsistent trials might weaken or disrupt cueing, as the violated expectation likely would reduce perceived agency. Passive/control trials, in which no volitional input would be involved, were expected to yield

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intermediate effect, reflecting neither enhanced agency nor explicit violation of expectations.

This approach inverts previous paradigms focused on how gaze influences agency [9],[29], offering instead a novel framework for understanding whether and how SoA can act as a top-down modulator of gaze-guided attention. The findings extend gaze cueing research into the domain of volitional control and predictive inference, highlighting the bidirectional interplay between internal states and social cues. Moreover, our work has broader implications: real-world joint attention and social coordination often depend on a person’s perceived role as an initiator or responder. Understanding how perceived action control modulates attention to gaze direction can enrich cognitive models of social perception, and have practical applications in education, therapy, and the design of socially responsive artificial agents.

Experiment

Method

Participants

Forty-three students participated in the experiment (7 males, age $M(SD)=24.7(8.4)$ years old). They either volunteered or participated in exchange for a course credit after giving their written informed consent. All participants had normal or corrected to normal vision.

Stimuli, Design, and Procedure

The design of the experiment was 2 (Gaze Cueing: congruent vs. incongruent) $\times$ 3 (Action–outcome consistency: consistent vs. inconsistent vs. control) within-subjects design. The face was generated by FaceGen program [11]. The face subtended around 13° of visual angle in height and 6° - in width, centered at the screen. From both sides of the face (5° to the nearest edge) squares were presented subtending 3.5° , positioned at the eye central ‘line’. Target colors were violet (BlueViolet, RGB: 138, 43, 226) and magenta (Magenta, RGB: 255, 0, 255) filled in one of the squares as soon as the eye movement was completed. Figure 1 presents a trial example.

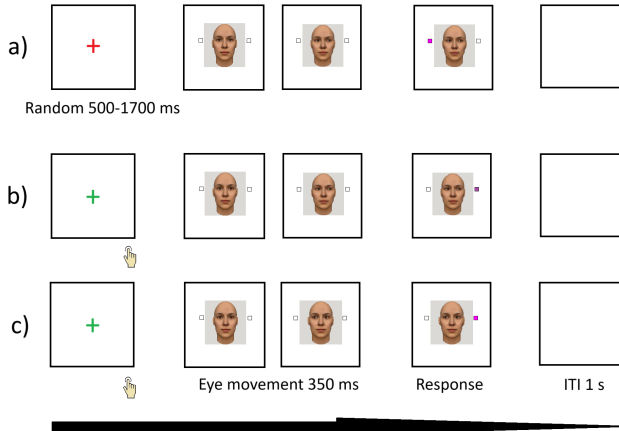


Figure 1: Trial timeline examples (not to scale). a) Control condition (passive expectation of face appearance), incongruent gaze cuing; b) and c) participants induce face appearance with a button press, choosing gaze direction, b) – congruent gaze cuing, c) – incongruent gaze cuing. During the response window participants discriminate between colors. ITI – intertrial interval.

Stimuli presentation and response collection was controlled by E-prime 2.0 software [7]. Participants were tested individually in a sound-proof booth in front of 22-inch monitor with resolution of 1920x1080 pixels and refresh rate of 50 Hz. They were positioned at around 60 cm from the screen. Participants were asked to perform two tasks: First task was to either pressing a button of their choice (left vs right) to initiate gaze movement or to refrain from pressing a button and to observe the gaze movements to the left or right. To initiate gaze movement participants were asked to press either button ‘6’ (right) or ‘4’ (left) on an extended keyboard: that is, the buttons were positioned on the right of the keyboard. The participants were asked to press these buttons using their right hand. The second task was color discrimination: as soon as the gaze movement was over, one of the squares positioned from both sides of the face was colored. Participants were asked to discriminate between colors (violet/magenta) as fast and as accurately as possible pressing a corresponding button on the computer keyboard (“a” or “d”) with their left hand. Response mapping was counterbalanced between participants. A trial started with a fixation cross. The participants were instructed to press a left or right button with their right hand (‘4’ or ‘6’) of their choice only if the cross was green. If a button press was not registered during 3 sec, then a green cross would appear again. After the button press, a face with two squares from each side of the face appeared. If the cross was red, participants would wait for a face to appear without pressing left/right button with their right hand. The red cross stayed on the screen for a random time between 500 and 1700 ms. Immediately after the face appearance the eyes moved either left or right, towards one of the squares. The movement took 350 ms. In case of

movement initiation by a participant, the eye movement could be either consistent with the intended by participant gaze direction or inconsistent. After the eyes reach their ultimate left or right position, one of the side white squares changed its color into violet or magenta. Participants had 1500 ms to respond using their left hand. The inter-trial interval was 1000 ms.

Participants first went through a practice block that consisted of 16 trials. The experimental procedure consisted of 168 trials (28 trials per condition). Trials were presented pseudo randomly so that there were no more than two consecutive trials with the same condition and color. The experiment took around 10-15 minutes to complete.

Results and Discussion

Data of ten participants were excluded due to low accuracy (<80%) or failure to follow the instruction. Data of two participants were excluded since they have complained about task difficulty. Finally, data of seven participants were excluded due to very long mean RT comparing to means of other participants (>700 ms). That is, we left data of only those participants for whom the color discrimination task was considerably easy. A post-hoc power analysis indicated that with the remaining sample size (24 participants) our experiment had sufficient power to detect even a small effect (power = 0.91 for an effect size $f = 0.25$). The post-hoc power analysis was conducted using G*Power 3.1.9.6 [12].

From the remaining data of 24 participants’ erroneous responses (7.1%) and outliers (5.06%) with RT outside of $\pm 2SD$ per participant and per condition means were excluded from the RT analysis. The data were averaged by subject and then entered into repeated measures ANOVA (rANOVA) with action-outcome consistency and gaze cuing congruency as within-subject variables. Corrected Bonferroni post-hoc test was applied where appropriate. Table 1 presents descriptive statistics per condition.

Action-Outcome	Gaze Cueing	
	Congruent	Incongruent
Consistent	546(78), 514–579	594(87), 558–631
Inconsistent	560(77), 528–593	583(86), 547–620
Control	586(92), 547–625	603(93), 564–643

Table 1: Mean RTs, standard deviations (SD) in parentheses, and 95% confidence intervals (CI) per condition, ms.

rANOVA obtained a significant main effect of gaze cuing ($F(1,23)=32.31$, $p<0.001$, $\eta_p^2=0.58$), suggesting that color discrimination was faster in congruent (564 ms) than in incongruent condition (594 ms). Main effect of action-outcome consistency was

also significant ($F(2,46)=5.96$, $p=0.005$, $\eta_p^2=0.21$). A post-hoc analysis showed that the control condition (595 ms) produced slower responses than the consistent (570 ms, $p=0.027$) and inconsistent (572 ms, $p=0.042$) conditions. There was no difference between consistent and inconsistent conditions ($p=0.776$). Most importantly, a significant two-way interaction ($F(2,46)=3.82$, $p=0.029$, $\eta_p^2=0.14$) was obtained (see Figure 2). A post-hoc comparison showed a significant gaze cuing congruence effect in action consistent condition (48 ms, difference between incongruent and congruent, $p=0.001$) as well as a congruence effect in inconsistent action-outcome condition (23 ms, $p=0.036$). Surprisingly, there were no significant congruence effect in the control condition (17 ms, $p=0.359$). In addition, Bonferroni post-hoc test found significant difference between consistent action-outcome and control ($p=0.039$) and incongruent control ($p<0.001$). The difference between the three mean conditions in incongruent gaze cue was insignificant (all $ps>0.8$) (cf. Figure 2).

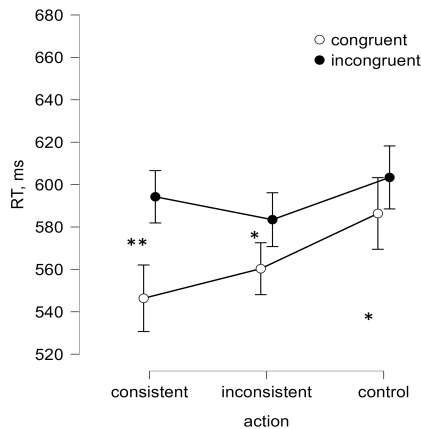


Figure 2: A gaze cuing by action consistency interaction. Vertical bars denote 95% CI. * $p<0.05$; ** $p<0.01$.

Overall, the results suggested that the gaze cuing was affected by action-outcome consistency. In particular, the consistent action-outcome condition made the effect two times larger than in inconsistent action-outcome condition (43 vs 23 ms). This result is in line with data suggesting that people tend to spare more attention towards actions they feel responsible for or in control which is linked to higher SoA ([34], [30]). The surprising lack of significant congruency effect in control condition partially can be explain by low expectation level and shift of attention. Since in the control condition eye movement started in a random time between 500 and 1700 ms after cross onset, participants would not be able to sustain their attention and would be then much less prepared to be ready in 350 ms for the color appearance. In contrast, in the action conditions, the sequence of events was quite fast, thus, making it easier to sustain the attention. To test this, we run a correlational analysis between time of participants button press (action decision, Mean=489 ms), random time of eye movement start in the control condition (Mean=1105 ms), and color discrimination time in all six

conditions. The result showed significant correlations between participants' action decision time and RTs of six conditions (r varied between 0.472 and 0.712, p varied between 0.020 and <0.001). In contrast, there was no correlation between six conditions' RTs and eye moving time (all $ps>0.8$). Thus, the correlation analyses confirmed our assumption. In addition, it was obvious that the task to discriminate between the two similar colors was quite difficult. Thus, probably, both possible reasons could be attributed to the lack of GCE in the control condition. On the other hand, note that the incongruent gaze cue was not affected at all. That is, it seems that only the gaze cue congruent condition was 'susceptible' to influence.

Overall, the results suggested that action control can modify GCE even though only in gaze cuing congruent condition, suggesting facilitation in comparison with the control. To test how the effect is distributed over time, distributional analyses of RT were conducted.

RT Distribution Analyses

RT distribution analyses were conducted on data tracing temporal dynamics of the gaze cuing effect, testing whether it occurs at earlier RTs and then decays or is developed over time. These analyses are important in that they can provide information on the level of penetrability of the effect given that the effect was influenced by action consistency.

To make the analyses clearer, the data were not reduced into the effect size as it is often done [22] and the conditions were preserved. Individual participants' data were rank-ordered per gaze cue congruence and action-outcome consistency and then divided into five bins of 20%. Mean RTs per subject and per condition were then computed per bin. Within-subject rANOVAs with factors gaze cuing (congruent vs. incongruent), action consistency (consistent vs. inconsistent vs. control) and bins (1-5) were conducted. To avoid the problem of multiple comparisons, the rANOVAs were run per action condition separately. Obviously, some results of the analyses mirrored the results presented in previous section. These were only reported and not commented. Since the key interest was the time distribution of the effect across bins and action-outcome consistency, the 3-way interactions were submitted to Bonferroni post-hoc test independently of significance of the interactions. Graphical representation of the results is shown in Figure 3.

Bin analysis of action-outcome consistency showed a significant main effect of gaze cuing ($F(1,9)=48.45$, $p<0.001$, $\eta_p^2=0.84$) and expected main effect of bins ($F(4,36)=357.79$, $p<0.001$, $\eta_p^2=0.98$). Post-hoc analysis showed that significant differences were present across all bins (all $ps<0.001$). The interaction between bins and gaze cuing was also significant ($F(4,36)=2.76$, $p=0.042$, $\eta_p^2=0.24$). The post-hoc test revealed differences between congruent and incongruent conditions across all bins (p varied from 0.022 to <0.001) but the fifth one ($p>0.5$) (cf. Figure 3).

Bin analysis of action-outcome inconsistency showed a significant main effect of gaze cuing ($F(1,7)=20.28$, $p=0.003$, $\eta_p^2=0.74$), main effect of bins ($F(4,28)=334.23$, $p<0.001$, $\eta_p^2=0.98$), and no interaction between bins and gaze cuing ($F(4,28)=1.20$, $p=0.331$,

$\eta_p^2=0.15$). Post-hoc analysis on bin main effect showed that, naturally, significant differences were present across all bins (all p s<0.001).

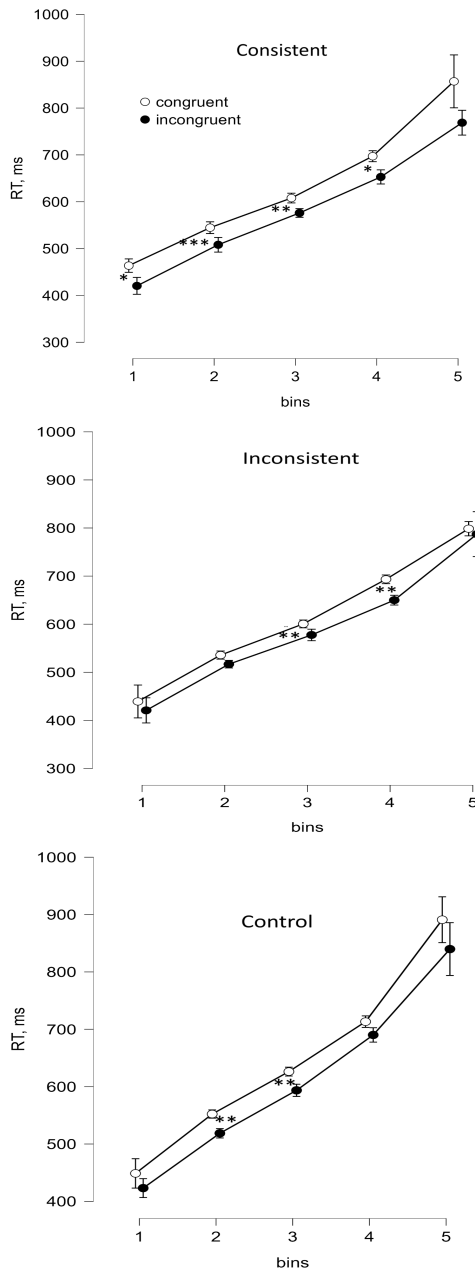


Figure 3: RT distribution across 5 bins, per action condition (consistent, inconsistent, control). Horizontal line represents bins from 1 to 5. Vertical bars denote 95% CI. * $p<0.05$; ** $p<0.01$; * $p<0.001$.**

The post-hoc test on the interaction revealed differences in bin 3 ($p=0.005$) and 4 ($p=0.005$) only. The difference in bin 1 and 5 was

highly insignificant ($p=1.0$), and in bin 2 close to significant level ($p=0.08$).

Bin analysis of control condition was very similar to the previous result: it showed a significant main effect of gaze cuing ($F(1,8)=23.43$, $p=0.001$, $\eta_p^2=0.75$), main effect of bins ($F(4,32)=835.92$, $p<0.001$, $\eta_p^2=0.99$), and no interaction between bins and gaze cuing ($F(4,32)=0.48$, $p=0.750$, $\eta_p^2=0.57$). Post-hoc analysis on bin main effect showed that significant differences were present across all bins (all p s<0.001). The post-hoc test on the highly insignificant interaction revealed differences only in bin 2 ($p=0.001$) and 3 ($p=0.002$). The differences in bin 1 and 5 ($p=1.0$) and in bin 4 ($p=0.341$) were highly insignificant.

Thus, very early impact of action-outcome consistency was observed in gaze cuing congruency effect, which was kept constant through the bins, disappearing in slow RTs (bin 5). At the same time, it was evident that action-outcome inconsistency as well as the control condition had quite weak overall GCE somewhere in the middle of RT distribution.

Taken together, the distribution analyses confirmed and extended the results of the experiment, showing that the GCE can be influenced by action-outcome consistency, as well as suggested that the effect itself may depend on task difficulty, predictability of stimulus appearance, and sustained attention, confirming previous findings that GCE combines both exogenous (reflexive, automatic) and endogenous (controlled, strategic) types of attentional orienting [35].

Conclusion

In conclusion, the present study adds the action control (SoA) as a potential modulatory factor in GCE, thereby supporting existing research that GCE is not merely an automatic, reflexive mechanism but a more complex phenomenon [24]. Our findings suggest that gaze cuing can be influenced by transient, experimentally induced internal states of the observer, such as action control.

Furthermore, the stark differences observed between the control condition and the action-outcome consistent and inconsistent conditions aligned with prior literature [1] and demonstrate that individuals preferentially allocate attention toward environmental elements they feel associated with or responsible for, reinforcing the idea that SoA enhances attentional engagement.

The relationship between SoA and gaze cuing is still in its early stages of empirical exploration. While the specific mechanism of this interaction requires further investigation, our findings confirm a possible existence of such a link which could be viewed as an important step towards clarifying it. Gaining deeper insight into this connection may contribute to a more integrated understanding of both phenomena, particularly SoA, which remains relatively underexplored.

Naturally, this study has its limitations. We did not include explicit or implicit SoA measurement scales, a decision made to streamline the experimental procedure and reduce cognitive load for participants. Nonetheless, we believe, based on a body of previous research [1] as well as research done in our lab (in preparation) that

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we can infer that self-agency was higher in conditions requiring participants to press a button to initiate gaze shifts. Our next experiments will incorporate SoA measurement tools to test this assumption directly.

Additional limitations stem from the demands of the dual-task paradigm. The speed requirement, as well as the subtle color discrimination between the two target hues, likely contributed to task difficulty. To further improve participant experience and make the task easier, we are considering sequential training for each subtask, as well as to introduce discrimination between basic colors (e.g., green and blue)

Looking ahead, follow-up studies may incorporate more extended measures of SoA, as well as a wider range of action modalities (e.g., full motor movements). This would allow for a more nuanced analysis of SoA and a clearer understanding of how it modulates social attentional orienting.

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